From raw listings to a monitored release

Design note / Project 4
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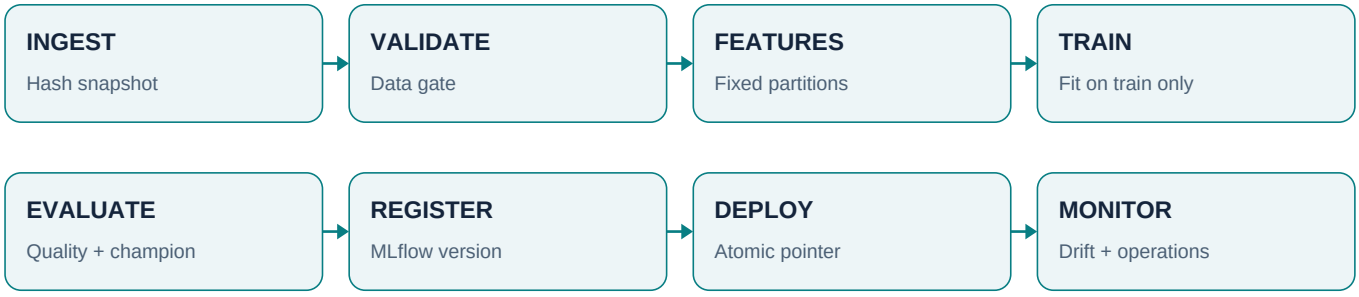
01 / Objective and measurable success

Predict property asking prices in INR lakhs for a real-estate analytics startup. Automate retraining and release while retaining the previous serving model whenever data or candidate quality is unacceptable. The supplied historical file has 4,000 listings; three later batches contain 500 labeled listings each.

Criterion	Configured objective and justification
Prediction quality	Validation MAE <=7 lakhs; RMSE <=9 lakhs. Initial Ridge results are 5.742 and 7.522; limits allow roughly 20% headroom while enforcing useful average error. These are project assumptions, not a client-signed SLA.
Safe promotion	Candidate MAE <=1.02 x current-model MAE on the same validation rows; absolute limits must also pass. First deployment has no incumbent comparison.
Operations	Warm p95 <=200 ms, request error fraction <=1%, model age <=30 days before a review alert. Network/container capacity still needs host verification.
Reproduction	Fixed seed 42, exact dependency lock, source/data hashes and split membership; metric tolerance 1e-6. Fresh-clone execution remains pending.

02 / Architecture mapped to the lifecycle



Execution order: top row, then bottom row. Failed gates keep the existing serving model.

FastAPI serves predictions and exposes health, lineage and Prometheus metrics. MLflow records parameters/configuration, scores, source artifacts, model versions and lifecycle labels. Prometheus/Grafana support operational observation; an HTML/JSON report covers the three labeled batches. The serving API reads model storage without write permission in Compose.

Tool decisions

A small persisted Python DAG substitutes for Prefect/ZenML: explicit dependencies, stage artifacts, reruns, hash verification, failure records and MLflow history. This keeps the local demonstration auditable; it is not a distributed scheduler. Git-tracked CSVs plus hashes/run snapshots replace DVC for the small supplied files. SciPy/pandas effect-size monitoring replaces Evidently. The brief permits justified equivalent free tools.

Gates, releases and evidence

03 / Data and evaluation protocol

One validated YAML defines paths, features, model settings and policy thresholds. Data gates require the declared schema, unique IDs, valid categories, plausible numeric ranges, >=100 rows and <=8% missingness per column. Only age, distance and neighborhood score may be missing; observed missingness is 4%. IDs and Batch never enter the feature matrix.

Historical split: 2,400 train / 800 validation / 800 final test, seed 42. Fit numeric imputation/scaling and categorical encoding only on training rows. Select Ridge by validation MAE, then evaluate the reserved test once. Its measured MAE is 5.775 lakhs and RMSE 7.506. Earlier whole-file summaries included test rows; prediction-error selection did not. Incoming batches are monitoring data, not training input.

04 / Release safety and traceability

Each stage persists its inputs/outputs and status. Rerunning an upstream stage invalidates descendants; artifact hashes prevent stale gate reuse. Promotion requires unchanged evaluated model bytes and unchanged incumbent version, then a finite-prediction check. A filesystem lock serializes updates. The pipeline copies an immutable model release and atomically switches production.json. Failed gates never change it.

MLflow uses validated/production/archived lifecycle tags and a champion alias, replacing deprecated stages. The API follows the serving manifest and retains a cached healthy model when replacement loading fails. Lineage links model hash, registry version, run ID, configuration, data hash and source snapshot; CI supplies a real commit ID once Git is enabled. Alias and file updates are not one transaction: an interrupted promotion requires reconciliation.

05 / Monitoring and response

Signal	Trigger / response
Input distribution	KS D >0.10 for numeric features; total variation >0.10 for categories; missing-rate change >3 percentage points. Investigate changes and schedule a candidate through all gates.
Labeled performance	MAE >7 lakhs or >1.25 x validation MAE. Batch 1: 5.562, no alert. Batch 2: 6.867, area alert only. Batch 3: 13.113, input and performance alerts. Retain current model until a candidate passes.
Service health	Observe latency, throughput, HTTP errors, predicted-price histogram, model age and load failures. Prometheus rules expose local alerts; outbound notifications are not configured.

06 / CI/CD, delivery status and limits

PR workflow runs tests and a lightweight pipeline. Main-branch workflow tests, smoke-checks a container, publishes a commit-tagged image and creates a release. Persistent deployment uses an opt-in self-hosted Docker runner. Model rollback was executed and verified. GitHub history, container execution/rollback, UI screenshots and the team recording remain external steps: no Git access, Docker runtime or server socket permission is available here.

Reference: MLflow Model Registry Workflows, https://www.mlflow.org/docs/latest/ml/model-registry/workflow/ . Full operational procedures and evidence: README.md, docs/OPERATIONS.md, reports/.